

Moving Tech AI Website

Firestore DB Schema - End to end explanation of the “companies” collection

Version	1.0
Collection	Companies
Total entries	150
Last Updated	03/21/2026

1. Collection Overview

The companies collection is the core data store of the Moving Software Platform. It holds structured, vendor-neutral profiles of software products used by moving and relocation companies worldwide. Each document represents one software product and contains identity fields, capability flags, coverage scores, status tracking, and a full audit trail.

Firestore path	/companies/{companyId}
Sub Collection path	/companies/{companyId}/audit_log
Public Read	Yes, all fields are publicly readable via the MVP
Write Access	Editors and Admins only via custom auth claims

NOTE: All automated updates (scraper results, change requests) pass through a human review step before being published. No field in this collection is ever overwritten by automation without editor approval.

2. Document ID Conventions

Each document is identified by a URL safe slug generated from the company name at import time. The slug is lowercase, with all non-alphanumeric characters replaced by hyphens.

Value	Meaning
Move-plnr	Move Plnr
Movedesk-crew	MoveDesk Crew
Smartmoving	SmartMoving
Moveman-job-moveman-pro	MoveMan Job (MoveMan Pro)

NOTE: If two companies produce the same slug, a numeric suffix is appended (example: acme-2). Slugs are immutable once created, renaming a company does not change the document ID.

3. Top Level Fields

These fields sit directly on the document root and describe the software product at a high level.

Field Path	Type	Required	Description
Name	String	Yes	Display name of the software product
Website	String Null	No	Primary marketing or product URL
Description	String Null	No	One sentence public facing summary shown on listing pages
Software_typology	String Null	No	Category label from the typologies

			reference collection
Target_market	String[]	No	Array of market segment enums (see Section 5)
Deployment_types	String[]	No	How the software is delivered (see Section 6)
Languages	String[]	No	Human language support, example: ["English", "Spanish"]
Keywords	String[]	No	Free form tags used for search and filtering
Notes_and_features	String Null	No	Full markdown notes for the detail page. Not shown on listing cards

4. Features Object

The features map groups capability flags into four functional domains. Each flag is a boolean where true = supported and false = not supported or null when the information is unavailable. Null is distinct from false because it means the data has not been verified, not that the feature is absent.

4.1 features.sales

Capabilities relevant to the sales and pre-move quoting workflow.

Field Path	Type	Required	Description
Features.sales.lead_management	Boolean Null	No	CRM style lead capture, tracking, and pipeline management
Features.sales.surveying	Boolean Null	No	Digital pre move survey and inventory capture tools
Features.sales.quoting	Boolean Null	No	Quote generation, pricing rules, and estimate delivery

4.2 features.move_management

Capabilities covering job booking through move execution.

Field Path	Type	Required	Description
Features.move_management.job_booking	Boolean Null	No	Booking confirmation, scheduling, and calendar management
features.move_management.move_mgmt	Boolean Null	No	Move workflow orchestration, job status, and coordination
features.move_management.quality	Boolean Null	No	Claims handling, damage reporting, and customer satisfaction tools

4.3 features.operations

Capabilities for field operations and logistics execution.

Field Path	Type	Required	Description
features.operations.dispatch	Boolean Null	No	Crew and vehicle dispatch, routing, and live updates
features.operations.shipment	Boolean Null	No	Shipment tracking, container management, or freight visibility
features.operations.crew_app	Boolean Null	No	Dedicated mobile app for on site crew (time tracking, inventory, signatures)
features.operations.storage	Boolean Null	No	Warehouse or self storage operational management

4.4 features.erp

Back office and accounting capabilities.

Field Path	Type	Required	Description
features.erp.accounts_receivable	Boolean Null	No	Invoicing, payment collection, and AR aging.
features.erp.accounts_payable	Boolean Null	No	Vendor bill management and AP processing.
features.erp.gl_banking_reporting	Boolean Null	No	General ledger, bank reconciliation, and financial reporting.
features.erp.hr	Boolean Null	No	Payroll, timesheets, and human resources management.

5. Coverage_Scores Object

Coverage scores are numeric summaries (0.0 to 1.0) expressing what percentage of features in each domain are supported. They are derived from the features flags and stored as floats for easy sorting and filtering.

Field Path	Type	Required	Description
Coverage_scores.overall	Number Null	No	Composite score across all four categories combined
Coverage_scores.sales	Number Null	No	Percentage of sales features marked true
Coverage_scores.move_management	Number Null	No	Percentage of move management features marked true
Coverage_scores.operations	Number Null	No	Percentage of operations features marked true
Coverage_scores.erp	Number Null	No	Percentage of ERP/back office features marked true

Example: a score of 0.75 means 3 out of 4 features in that domain are supported. Scores are null when all underlying flags are null (unverified data).

NOTE: coverage_scores.overall is composite indexed with status.review_status to support sorting the public listing by completeness score.

6. Enum Values

6.1

One or more of the following values appear in the target_market array.

Value	Meaning
Domestic_local	Software targeting local or domestic household moves
International	Software supporting international relocation shipment.
Relocation_management	Corporate RMC platforms managing assignee relocations
Military	Platforms specialized for military PCS moves
Fine_art	Platforms for high value fine art and specialty transport
All_movers	General purpose tools suitable across all move types

6.2

One or more of the following values appear in the `deployment_types` array.

Value	Meaning
Web	Browser based SaaS application, no installation required
Cloud	Cloud hosted platform (may have desktop or API access)
Native_mobile	Native iOS or Android application
On_premise	Installed and ran on the customer's own servers
hosted	Vendor hosted but may offer on premise option

7. Status Object

The status object tracks where a company profile is in the editorial and vendor confirmation workflow. It controls what is shown publicly and what requires further review.

Field Path	Type	Required	Description
Status.review_status	String	Yes	Editorial state of the profile. (See ENUM table below)
Status.vendor_confirmed	Boolean	Yes	True if the vendor has reviewed and confirmed the profile data

Review_status ENUM

Value	Meaning
Draft	Profile exists but has not been through any internal review. Not shown publicly
Unverified	Profile has been imported or edited but awaits internal review. Shown publicly with an 'Unverified' badge
Verified	Profile has been reviewed internally and/or confirmed by the vendor. Shown publicly without caveats
Inactive	Software is discontinued, acquired, or otherwise no longer active. Hidden from public listings

8. Meta Object

The meta object stores creation and timestamp information. All timestamps are Firestore server timestamps (UTC).

Field Path	Type	Required	Description
Meta.created_at	Timestamp	Yes	When the document was first created in Firestore
Meta.updated_at	Timestamp	Yes	Last time any field on the document was modified. Auto updated by the onCompanyWrite Cloud Function
Meta.last_scraped_at	Timestamp Null	No	Last time the automated scraper successfully fetched this company's

			website. Null if never scraped
Meta.data_source	String	Yes	Origin of the most recent write. Values: initial_import, admin_edit, vendor_confirmed, automated_scrape

9. Audit_Log sub-collection

Every write to a company document triggers the onCompanyWrite Cloud Function, which appends an immutable entry to the audit_log sub-collection at:

/companies/{companyId}/audit_log/{logId}

Field Path	Type	Required	Description
Change_type	String	Yes	One of: created, updated, deleted
Before	Object Null	Yes	Full document snapshot before the change. Null for created events
After	Object Null	Yes	Full document snapshot after the change. Null for deleted events
Changed_at	Timestamp	Yes	Server timestamp of the write
Changed_by	String	Yes	UID of the authenticated user, or 'system' for automated writes

NOTE: Firestore security rules block update and delete operations on audit_log documents. Once written, an audit entry cannot be altered. This ensures a complete and tamper proof change history for every profile.

10. Indexes

The following composite indexes are deployed to firestore.indexes.json and support the filtering and sorting patterns used by the public listing and the admin dashboard.

Fields	Order	Use Case
status.review_status + meta.updated_at	ASC + DESC	Admin dashboard, latest updated profiles
software_typology + status.review_status	ASC + ASC	Filter listing by software category
target_market[] + status.review_status	CONTAINS + ASC	Filter listing by move type
deployment_types[] + status.review_status	CONTAINS + ASC	Filter listing by deployment
target_market[] + software_typology	CONTAINS + ASC	Combined market + category filter
status.vendor_confirmed + meta.updated_at	ASC + DESC	Admin, show unconfirmed profiles
coverage_scores.overall + status.review_status	DESC + ASC	Sort listing by coverage score

11. Security Rules

Access is governed by custom JWT claims set on authenticated users. Three roles are defined: admin (full access), editor (read and write companies, manage change requests), and unauthenticated users (public read only).

12. Example Entry

Document ID: "smartmoving"

```
name: "SmartMoving"
website: "https://www.smartmoving.com"
description: "All-in-one CRM and operations platform for moving companies."
software_typology: "Moving ERP / CRM"
target_market: ["domestic_local", "all_movers"]
deployment_types: ["web", "cloud"]
languages: ["English"]
keywords: ["CRM", "Dispatch", "Quoting", "Invoicing"]
```

features:

sales:

lead_management: true

surveying: true

quoting: true

move_management:

job_booking: true

move_mgmt: true

quality: true

operations:

dispatch: true

shipments: false

crew_app: true

storage: false

erp:

```
accounts_receivable: true
accounts_payable:    false
gl_banking_reporting: true
hr:                  false
```

coverage_scores:

```
overall:            0.75
sales:              1.00
move_management:    1.00
operations:         0.50
erp:                0.50
```

status:

```
review_status:      "verified"
vendor_confirmed:    true
```

meta:

```
created_at:         2025-01-15T10:00:00Z (Timestamp)
updated_at:         2025-06-01T14:32:00Z (Timestamp)
last_scraped_at:    2025-07-14T02:00:00Z (Timestamp)
data_source:        "vendor_confirmed"
```


13. Potential Future Fields

The following fields are out of scope for the MVP but are reserved for future phases to avoid breaking schema changes.

Field Path	Type	Required	Description
integrations	string[]	Future	List of integration partners (example: QuickBooks, Salesforce)
ai_capabilities	object	Future	Structured AI feature flags (example: ai_inventory, ai_quoting, voice_agent)
pricing_model	string	Future	One of: subscription, per_user, per_job, custom
company_size_target	string[]	Future	Target customer size: small, mid_market, enterprise
functional_score	number null	Future	Weighted coverage score incorporating importance weights per feature
reviews_count	number	Future	Aggregate count of user reviews (from reviews sub collection)
reviews_avg_rating	number null	Future	Average star rating across all published reviews

